



PolySim OS

The Solver Catalog

173 interactive simulations with real numerics — no install, no cloud, free forever.
The Wolfram Alpha of interactive simulation

Flagship Engines, Fields & GPU Compute (19)

Visual Node Graph

Wire blocks into a live dataflow — the real editor. Symbolic calculus built in.

2D Fluid (CFD)

Interactive incompressible Navier–Stokes via Stam's stable-fluids method.

Heat & Wave Fields

2D heat diffusion and the 1D wave equation, solved live.

AI Surrogate

Train an ML surrogate on our solver for instant predictions. The PhysicsX play.

GPU Compute (WebGPU)

Hundreds of thousands of particles on your GPU via WGSL shaders.

WebGPU Fluid

High-res GPU smoke advection you can stir.

GPU N-Body

Thousands of bodies, full $O(n^2)$ gravity on the GPU.

GPU PDE Solver

Steady heat/Poisson via GPU Jacobi at high res.

GPU 3D Fluid

3D smoke, simulated + raymarched on the GPU.

Particle-Mesh N-Body

100k+ bodies via GPU particle-mesh gravity.

3D CFD

3D Navier–Stokes plume with z-slices.

3D N-Body

Gravitation in 3D with a drag-to-orbit camera.

3D FEA Space Frame

3D structural analysis; orbit the deformed tower.

3D Heat Diffusion

Volumetric heat with slices and orbit view.

FEA Truss

Finite-element structural analysis: forces & deflection.

Meshing + BCs

Paint boundary conditions; solve steady heat.

Optimize + UQ

Gradient descent and Monte-Carlo uncertainty.

RLC Circuit

Step response and damping regimes.

Gradient Descent

Optimization on loss landscapes.

Physics, Mechanics & Thermodynamics (25)**Particle / N-Body**

Gravity, orbits, and impulse collisions with a real symplectic integrator.

Electrostatics

Charges, potential heatmaps, and field lines.

Double Pendulum

The textbook chaotic system, integrated with RK4.

Projectile Motion

Ballistics with air drag; tune angle and speed.

Ising Model

A live magnetic phase transition via Monte Carlo.

Strange Attractors

Lorenz, Rössler, Aizawa & more in 3D.

Wave Interference

Two-source ripple-tank fringes.

Random Walk

Brownian motion and diffusion.

Kepler Orbits

Conic-section orbits under gravity.

Double-Slit

Quantum interference + diffraction.

Cloth / Spring-Mass

Verlet cloth you can grab and swing.

Gravity Well

Curved-spacetime rubber sheet + orbit.

Magnetic Pendulum

Fractal basins of attraction.

Percolation

A phase transition in connectivity.

DLA Growth

Branching fractals from random walks.

Ray Optics / Lenses

Trace rays through lenses.

Gas in a Box

Kinetic theory → pressure ($PV=nRT$).

Elastic Collisions

Momentum & energy conservation.

Buoyancy

Archimedes — float or sink by density.

Rocket Equation

Tsiolkovsky Δv budget calculator.

Blackbody Radiation

Planck curve + Wien's law.

Pendulum Wave

Phase art from many periods.

Orbital Transfer

Hohmann transfer Δv .

Standing Waves

Harmonics, nodes & antinodes.

Doppler Effect

Moving-source wavefronts & sonic boom.

Waves, Optics & Acoustics ⁽⁶⁾

Snell's Law

Refraction & total internal reflection.

Diffraction Grating

N-slit interference maxima.

Beats

Two close tones interfere.

Curved Mirrors

Concave/convex ray tracing.

Prism Dispersion

White light into a spectrum.

Fourier Transform

Signal $\rightarrow$ frequency spectrum.

Mathematics (22)**Dynamical Systems**

Lorenz, SIR epidemics, pendulums, predator–prey, and reaction–diffusion.

Symbolic Math

Parse, differentiate, simplify, solve, and plot — a real CAS in your browser.

Vector Fields

Plot any $F(x,y)$ as a live quiver diagram.

Notebook

Cells of prose + symbolic math + compute.

Fractal Explorer

Zoom into the Mandelbrot and Julia sets.

Fourier Series

Build waves from sine harmonics.

Function Grapher

Plot up to three functions at once.

3D Surface Plotter

Orbit any $z = f(x, y)$ surface.

Matrix Calculator

Multiply, invert, determinant, transpose.

Cellular Automata

Wolfram rules & Conway's Life.

Taylor Series

Polynomial approximation, live.

Newton's Method

Root finding, tangent by tangent.

Distributions

Normal, Poisson, binomial & more.

Complex Functions

Domain coloring of $f(z)$.

Lissajous Curves

Two sine waves, endless patterns.

Bifurcation Diagram

Logistic map route to chaos.

Direction Field

Slope fields + solution curves.

Riemann Sums

Integrals as stacked rectangles.

Eigenvectors

2×2 transforms & invariant directions.

Markov Chains

Converge to a stationary distribution.

Parametric Grapher

Trace $x(t)$, $y(t)$ curves.

Euler vs RK4

ODE integration accuracy.

Statistics & Data Science ⁽⁸⁾

Hypothesis Test

p-values & rejection regions.

Linear Regression

Least squares & R^2 .

Central Limit Theorem

Means go normal.

Confidence Intervals

What 95% really means.

Bayesian Inference

Prior $\times$ likelihood $\rightarrow$ posterior.

Bootstrap

Resampling confidence.

PCA

Axes of maximum variance.

A/B Test

Two-proportion significance.

Chemistry & Materials (8)**Molecular Dynamics**

Lennard-Jones atoms; melt a lattice.

Titration Curve

pH vs base added; equivalence point.

Reaction Kinetics

Rate laws & Arrhenius decay.

Chemical Equilibrium

Le Chatelier shifts.

pH Calculator

pH/pOH and the acid-base scale.

Radioactive Decay

Half-life exponential decay.

Maxwell-Boltzmann

Molecular speed distribution.

Ideal Gas Law

$PV = nRT$ with a live piston.

Computer Science & AI (10)

Sorting Visualizer

Watch algorithms sort in real time.

Boids Flocking

Emergent flocking from three rules.

A* Pathfinding

Shortest path, draw walls.

Maze Generator

Carve & solve perfect mazes.

Neural Network

Train an MLP, watch it learn.

k-Means Clustering

Unsupervised clustering, live.

Convex Hull

Wrap points in $O(n \log n)$.

L-System Fractals

Grow plants from grammars.

Turing Machine

The model of computation.

Image Convolution

Kernels behind every CNN.

Complex Systems & Chaos (9)

Traffic Flow

Phantom jams from random braking.

Forest Fire Model

Self-organized criticality.

Abelian Sandpile

Avalanches build a fractal.

Schelling Segregation

Mild bias, sharp segregation.

Langton's Ant

Two rules, emergent highway.

Reaction-Diffusion

Gray-Scott Turing patterns.

Elementary CA

Wolfram's 256 rules.

Genetic Algorithm

Evolve toward the peak.

Ant Colony Optimization

Swarm-solved TSP.

Biology & Medicine (10)

Epidemic Network

Agent-based SIR on a contact graph.

Spatial Predator-Prey

Agent-based ecology with waves.

SIR Epidemic

Outbreak compartment model.

Hodgkin-Huxley Neuron

Action potential spikes.

Lotka-Volterra

Predator-prey cycles.

Hardy-Weinberg

Allele frequency & selection.

Enzyme Kinetics

Michaelis-Menten curves.

Logistic Growth

Population carrying capacity.

Genetic Drift

Wright-Fisher allele drift.

Sequence Alignment

Needleman-Wunsch DNA.

Astronomy & Space (8)**Exoplanet Transit**

Light-curve transit method.

H-R Diagram

The map of the stars.

Lagrange Points

Restricted 3-body equilibria.

Roche Limit

When tides shred a moon.

Hubble's Law

The expanding universe.

Telescope Optics

Resolution & magnification.

Stellar Parallax

Distance from angle shift.

Escape Velocity

Speed to leave a world.

Earth, Climate & Environment (8)

Energy Balance

Greenhouse equilibrium temp.

Daisyworld

Planetary self-regulation.

Milankovitch Cycles

Orbital forcing of ice ages.

Tsunami Propagation

Shallow-water wave shoaling.

Carbon Cycle

Reservoir box model.

Earthquake Shaking

Magnitude, energy, intensity.

Groundwater Drawdown

Well cone of depression.

Lapse Rate

Atmosphere & cloud base.

Finance & Quantitative (8)

Black-Scholes

Option pricing + Greeks.

Option Payoff

Strategy P/L diagrams.

Monte Carlo (GBM)

Price-path distributions.

Efficient Frontier

Portfolio risk vs return.

Value at Risk

VaR & expected shortfall.

Bond Pricing

Price, yield & duration.

Compound Interest

Investment growth over time.

Loan Amortization

Mortgage payment breakdown.

First Responders (Fire | Police | EMS | Hazmat) (8)**Wildfire Spread**

Wind + slope fire behavior.

Hazmat Plume

Gaussian dispersion + PAD.

Building Evacuation

Egress flow & clear time.

START Triage

Mass-casualty triage tool.

Fire Hose Hydraulics

Friction loss & pump pressure.

Skid-to-Stop Speed

Accident reconstruction.

Radio Range

VHF/UHF link budget.

Blast Standoff

Overpressure & safe distance.

Structural & Civil Engineering ⁽⁸⁾

Beam Deflection

Sag & bending moment.

Column Buckling

Euler critical load.

Mohr's Circle

2D stress transformation.

Shear & Moment

SFD/BMD diagrams.

Retaining Wall

Rankine earth pressure.

Seismic Base Shear

Equivalent lateral force.

Soil Bearing

Terzaghi capacity.

Concrete Beam

RC flexural capacity.

Electrical & Signal Processing ⁽⁸⁾

Bode Plot

Frequency response.

Filter Designer

RC/RLC filter response.

PID Controller

Step-response tuning.

Transmission Line

VSWR & reflections.

Op-Amp Circuits

Amplifier configurations.

Three-Phase Power

Grid power & phasors.

Transistor Bias

BJT Q-point load line.

Sampling & Aliasing

The Nyquist theorem.

Robotics, Control & Kinematics (8)**Forward Kinematics**

Joint angles → end effector.

Inverse Kinematics

Target → joint angles.

Differential Drive

Wheel speeds → path.

Cart-Pole Balance

Inverted pendulum control.

Quadcopter Control

Nested PID flight.

RRT Path Planning

Sampling-based planner.

DC Motor

Speed-torque & step.

Kalman Filter

Optimal state estimation.